

Tacit: Confidential Assets and DeFi on Bitcoin, Bridged by Zero-Knowledge Reflection

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2026

Abstract

Bitcoin can hold value without a trusted party, but issuing assets on it, hiding amounts, or trading them has meant accepting one: an off-chain proof courier, a federation, a rollup operator, or a custodian. Tacit removes that party in two steps.

First, it is an indexer-validated metaprotocol. Assets live in Taproot envelopes, amounts are Pedersen commitments with range proofs, and conservation is a Schnorr signature over the excess. Any indexer running the specification reaches the same state from the chain alone.

Second, Tacit reflects that state into an immutable confidential pool on Ethereum. Every pool transition is one SP1 zero-knowledge proof. Bitcoin blocks reach the pool through a full-proof-of-work header relay and an SP1 guest that folds Tacit envelopes. Pool state returns to Bitcoin through a light-client proof verified recursively inside the same guest. No signer stands between the chains.

On this base the pool runs confidential DeFi:

- AMM swaps and routes, with a prover-blind batch op backed by a ceremony Groth16 circuit;
- OTC trades, bids and adaptor-signature swaps;
- stealth payments;
- a CDP issuing cUSD;
- cBTC backed by self-custody BTC locks;
- farms;
- gasless relaying with the fee bound inside the proof.

Every balance recovers from a private key and public chain data. Bytes are reserved for constructions that Bitcoin covenants would make fully trustless.

1 Introduction

A useful asset layer on Bitcoin needs five properties:

1. its state is public data anyone can check;
2. amounts are hidden;
3. value can move to where computation is cheap without trusting a bridge;
4. BTC itself can back a fungible asset;
5. a user can recover everything from a key.

Existing designs each give up one:

Design	What it gives up
Ordinals, Runes	Amounts are public.
RGB, Taproot Assets	Validity moves off-chain, so losing a proof means losing the balance.
Liquid	Relies on a federation.
Rollups and Bitcoin L2s	Rely on an operator set.
Wrapped BTC	Relies on a custodian or a threshold group.

Tacit provides all five by composing existing systems, not by adding consensus. One gap remains: cBTC custody is economic until Bitcoin has covenants (§6, §11). Bitcoin orders the data, and indexers read it (§2). Commitments hide amounts (§3). An immutable contract on Ethereum accepts only state transitions that come with a zero-knowledge proof (§4). The two chains learn about each other through proofs, not signatures (§5).

2 A metaprotocol on Bitcoin

A Tacit op is a commit/reveal pair. The commit pays to a Taproot output whose only script leaf is:

$\langle P \rangle$ OP_CHECKSIG OP_FALSE OP_IF "TACIT" 0x01 $\langle payload \rangle$ OP_ENDIF

The reveal spends the commit by script path, so the payload rides in witness data. Bitcoin nodes do not interpret it. Indexers do: a Tacit UTXO is valid iff the op that created it passes its rules and each of its asset inputs is itself valid.

The rules are deterministic, so two indexers that see the same bytes reach the same verdict. There is no quorum, no vote and no leader. Ordinals and Runes use the same trust model. What users trust is the published specification and its open reference implementations.

The Bitcoin reflection guest (§5) is a second, independent check. It re-implements the rules for every op it folds, and a verifying key that cannot be changed fixes its verdict.

Ops cover:

- **Issuance:** confidential supply, or fair-launch mints against a public cap. TAC, the native asset, was issued with T_CETCH and a zero mint authority, so its supply is fixed.
- **Transfers:** confidential.
- **Trades:** atomic OTC trades against BTC, bids that a watchtower fills while the buyer is offline, and a native AMM with per-trade swaps, routes, uniform-price batch clearing and LP farms.
- **cBTC:** locks and redeems.
- **Cross-chain:** bridge burns, re-mints of notes crossing back from Ethereum, and value-free calls in both directions.

An unknown opcode creates no state, so the protocol grows by adding opcodes.

3 Confidential value

Every amount is a Pedersen commitment on secp256k1:

$$C = v \cdot H + r \cdot G,$$

where G is Bitcoin's generator and H is a nothing-up-my-sleeve point derived from $\text{SHA256}(\text{"tacit-generator-H-v})$. Commitments hide the value perfectly and add homomorphically. A transaction balances iff its excess

$$E = \sum C_{\text{out}} + b \cdot H - \sum C_{\text{in}}$$

carries no H component, where b is any public burn. Tacit proves this with a BIP-340 signature under E , a Mimblewimble-style kernel. Only someone who knows every blinding can sign, and nobody can sign for an excess containing value.

Range proofs bound every output to $[0, 2^{64})$, aggregated over up to eight outputs. New transfers use Bulletproofs+, and classic Bulletproofs remain accepted. The recipient's blinding and an 8-byte amount keystream come from an ECDH shared secret between sender and recipient, anchored to the transaction's first asset input. A recipient therefore finds and opens every credit from its own key and the chain: no share link, no sync server.

The same commitment, with the same G and H , is the note in the Ethereum pool. One note format on both chains lets value cross between them without changing representation.

4 The confidential pool

The pool is one immutable contract holding three depth-32 keccak Merkle trees: notes, locks and CDP positions. It also holds three spent sets, an escrow for every wrapped asset, and the reflected state of Bitcoin. It has no owner and no pause.

A note is $(\text{asset}, C, \text{owner})$. Its leaf is a keccak hash of those fields, and its nullifier is derived from the leaf. For an owned note the nullifier also takes the owner's secret nullifier key, so an observer cannot tell which leaf a spend consumes. Bearer and Bitcoin-homed notes use a nullifier derived from the leaf alone, which anyone can compute.

Every state change is a batch of up to 256 ops proven by one SP1 program. Per op, the program checks:

- that each input leaf is a member of a known root, without revealing which leaf;
- the range proofs;
- the kernel or opening proofs;
- the op's own rules.

It then commits the public effects: nullifiers, new leaves, payouts, fees, swaps and liquidity changes, bridge claims and memos. The contract verifies the proof against a pinned verifying key and applies exactly those effects.

A spend proves membership against any root the tree has ever had. The anonymity set of an owned-note spend is therefore every owned note of its asset.

Nothing in the pool accumulates cost with use:

- an insert is 32 hashes at any fill;
- nullifier and root sets are constant-time maps that are never iterated;
- the guest touches history only through fixed-depth proofs.

The pool binds every proof to $\text{CHAIN_BINDING} = \text{keccak}(\text{chainid} \parallel \text{address})$, so a proof for one deployment is invalid in any other.

5 Reflection: a bridge without signers

Bitcoin to Ethereum. Anyone may extend `BitcoinLightRelay`, a header relay that checks full proof-of-work and follows the heaviest chain. The reflection guest proves a run of blocks on top of it. For each block it:

1. re-hashes every transaction to the header’s Merkle root;
2. nullifies every input that spends a reflected note;
3. re-verifies and folds each Tacit envelope it understands, and skips the rest;
4. extends a digest chain over the result.

The pool accepts an attestation only if its prior digest matches, its tip is buried at least 24 blocks under the relay tip, and its counters match the pool’s own. The pool then holds current roots of Bitcoin’s Tacit notes, of its spent set and of its bridge burns, plus cBTC lock records.

Value enters the pool from Bitcoin in two ways:

- **Bridge burn.** The note is destroyed on Bitcoin under a burn id that commits to the target deployment, and the pool mints it exactly once.
- **Fast lane.** A Bitcoin-homed note bound to this deployment (by `T_CXFER_BOUND`) is spent directly on Ethereum. The proof must show that its nullifier is absent from the reflected Bitcoin spent set, and it must carry a signature from the note’s Bitcoin key. The pool records the consumed source so that reflection retires the note on Bitcoin.

Ethereum to Bitcoin. A second guest, built on the `sp1-helios` sync-committee light client, proves finalized pool storage:

- cross-out commitments,
- consumed sources,
- outbox messages.

The committee chains from period to period, with each step committed. The Bitcoin reflection guest verifies this proof recursively and folds, by membership:

- re-mints of crossed-out notes on Bitcoin;
- outbox messages;
- retirements of fast-lane notes.

Calls. A value-free Bitcoin envelope can authorize an Ethereum call, which an executor contract runs once it is reflected. Ethereum can post messages that reflection delivers to Bitcoin.

The bridge’s soundness rests on Bitcoin proof-of-work, the Ethereum sync committee and SP1. There is no multisig, attester set or wrapped-asset IOU. Its liveness needs someone to run the provers. Anyone may, and Ethereum-homed notes stay spendable if nobody does.

6 Confidential DeFi

Swaps. Pools are constant-product curves with a fee tier and an optional protocol-fee switch. Reserves are public. A plaintext AMM contract, `TacitPublicAmm`, trades against the same reserves, so public and confidential liquidity share one curve.

- `OP_SWAP_ROUTE` swaps a hidden input note through up to four pools, with an end-to-end minimum output. The dapp settles every swap this way; a single swap is a one-hop route.

- **OP_SWAP** clears many hidden-output intents at one uniform price. Traders in one batch pay the same price, so they cannot sandwich each other. The dapp offers it as an opt-in batch.

Each swap changes public reserves, so a swap settled alone shows its size. A batch shows only its net change.

Prover-blind swaps. Whoever proves an **OP_SWAP** or **OP_SWAP_ROUTE** sees its amounts. **OP_SWAP_BLIND** keeps them out of the SP1 witness. The batcher that computes the clearing and produces its Groth16 proof still sees them; the SP1 prover and the chain do not. The op is enabled in the deployed guest, but no client emits it yet.

Each trader commits to its input on BabyJubJub, the curve native to BN254. A 169-byte cross-curve sigma proves that commitment equals the trader’s secp256k1 note. The batch carries one Groth16 proof of the **amm_swap_batch** circuit, which proves over hidden amounts that:

- the clearing price is correct;
- every fill is correct;
- every minimum output is met.

The circuit’s verifying key comes from the AMM trusted-setup ceremony: Hermez Phase 1, then a Phase 2 of 5,018 contributions sealed by a Bitcoin-block beacon. It is compiled into both SP1 guests, which verify the pairing in-guest. The reflection guest uses the same key for **T_SWAP_BATCH** on Bitcoin.

A circuit fits here because uniform-price clearing over hidden amounts is cheap as a circuit and expensive as a zkVM program. The guests’ own verifying keys fix the circuit key.

Trades without a pool.

- **OP_OTC** swaps two parties’ notes directly.
- **OP_BID** is a limit order with a pre-signed grid of partial fills, so the buyer need not be online.
- Adaptor locks tie a claim to the revelation of a Schnorr signature scalar, which gives atomic swaps against the other chain.

Stealth payments. A payer locks a note under a one-time key derived from the recipient’s published key. The recipient claims it with a signature, and the payer may refund after a deadline. A Bitcoin bridge burn can mint straight into such a lock.

cUSD. A CDP locks cBTC collateral and mints cUSD. The position’s owner cannot be linked to it. Its amounts are public, because the controller prices them.

- Minting requires 150% collateral today, and a position can be liquidated below 130%.
- Liquidation burns the full accrued debt and seizes the basket.
- A stability fee and a savings rate are built in and dormant.
- Policy lives in the **CollateralEngine**, which the pool consults only to accept or reject. The ops multisig governs it. Its ratios and fee are capped on-chain, and any change against borrowers takes effect only after notice.

cBTC. A user locks BTC on Bitcoin with a **T_CBTC_LOCK** envelope, which pre-commits to the note it will mint. Reflection records the lock’s value, and the pool mints cBTC once per lock. The locker holds the lock’s key. A wstETH escrow of at least the engine’s escrow ratio times the lock value ($1.5\times$ today) makes spending it unprofitable:

- a spend without a matching redeem is visible to reflection, and anyone can slash the escrow;

- a proven redeem returns the escrow.

cBTC supply is bounded by reflected locks, with no oracle in the mint path. Custody is economic rather than enforced. Covenants would close this gap (§11).

Farms. LP-share notes bond into a receipt note, and the farm controller records the reward-per-share at entry. A harvest pays the accrued reward as a note, drawn from the controller’s escrowed treasury. Unbonding returns the shares. The live farms pay wTAC, a 1:1 wrapper of TAC.

7 Gasless and self-sovereign

Any op may be relayed. The relayer’s fee is part of what the proof commits to, bound by the conservation kernel, the opening proof, or the note’s spend signature. The relayer pays gas and collects exactly that fee. It cannot redirect a payout, raise the fee, or submit the proof after its deadline. A fee is zero or has at most two significant digits.

Once value is in the pool, a user never needs ETH to move it.

The relayer is optional. `settle` is open to anyone, proofs can be generated locally or on an open prover network, and the three guest programs rebuild byte for byte from source. A user can prove locally, settle directly, and involve no third party.

8 Immutable deployments, opt-in succession

The pool, its guests and their verifying keys cannot be upgraded. The protocol changes by deploying a successor pool that users choose to enter.

A pool has one privileged governance call, `createNextGen`, which deploys its successor once and records it. The ops multisig holds it. The pool’s only other privileged callers are the public AMM and the farm controller, and neither can move a user’s note or escrow.

After succession the old pool refuses new value. Every exit and every release of value already committed stays open, as does its reflection. The successor resumes the shared Bitcoin lane from the predecessor’s handoff record or current attested state, so reflection has no gap. Users move by exiting one pool and entering the next. Nothing moves value on their behalf, and a pool deployed by anyone else is an isolated system that cannot touch the lineage.

9 Privacy

	Hidden	Public
Bitcoin	amounts; stealth recipients	addresses, the transaction graph, asset ids, burn amounts
Pool	amounts; which owned note a spend consumes	the deposit and withdrawal boundary, AMM reserves and their changes, CDP amounts
Relay	your key, seed and blindings	the witness of the ops it proves

The pool’s anonymity set is every owned note of an asset, not one denomination. `OP_SWAP_BLIND` would also hide trade sizes from the SP1 prover, but no client emits it yet. A user who wants no

one to see a witness proves locally.

10 Trust and limits

Relied on:

- Bitcoin and Ethereum consensus;
- the soundness of SP1 and of Groth16;
- the spl-helios sync committee;
- for the circuits in §6 only, at least one honest contributor to each ceremony;
- cUSD's oracle;
- the ops multisig (2-of-4, with a built-in one-hour delay), within the on-chain bounds of the collateral and farm controllers.

Not relied on: relayers, the hosted API, gateways and explorers. The reference indexer reads Bitcoin from public Esplora endpoints with fallbacks, and any other source gives the same state.

Stated limits:

- Reflection halts, rather than rewrites, on a Bitcoin reorg deeper than its confirmation depth.
- Anyone can submit a relayed proof first and collect its fee. The user's outcome does not change. This follows from a proof that works for any relayer.
- cBTC is economically secured, not custodially guaranteed.
- On-chain Bitcoin transfers carry a large witness. Tacit on Bitcoin is suited to settlement, and the pool is where frequent activity belongs.

11 Bitcoin covenants and the road ahead

Several constructions would be fully trustless if Bitcoin could restrict where an output may be spent. Tacit reserves room for them now:

- **Covenant cBTC.** Pool op 5, OP_COVENANT_MINT, is held for cBTC minted against a lock whose sats can leave only through a redemption. It would reuse the existing mint's value binding with no escrow, so cBTC would need no economic assumption.
- **On-chain bid escrow.** A covenant could bind a buyer's pre-signed input to its bid template. That would replace the watchtower that fills today's offline bids, and it would activate the reserved batch-fill and both-sides match bytes.
- **Fractional slots.** Reserved opcodes would split a BTC slot into fungible shares and recombine them, with no shared vault.

Any of CTV, OP_CAT, OP_CHECKSIGFROMSTACK, OP_VAULT or an equivalent suffices. Each would ship as additive opcodes, or as a new pool deployment users opt into. None changes what is already live.

Cryptography changes the same way: a better range proof is added alongside the old one, as Bulletproofs+ was. Proof systems verifiable in Bitcoin script could move parts of indexer validation into consensus.

12 Conclusion

Tacit builds an asset layer on Bitcoin without a bridge signer or custodian:

- a deterministic metaprotocol with hidden amounts;
- an immutable, proof-gated pool that carries the same notes into confidential DeFi;
- a bridge made of proofs in both directions.

Where cryptography can enforce a property, it does. Where Bitcoin cannot yet enforce one, the gap is stated, bounded, and reserved for a covenant to close.

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